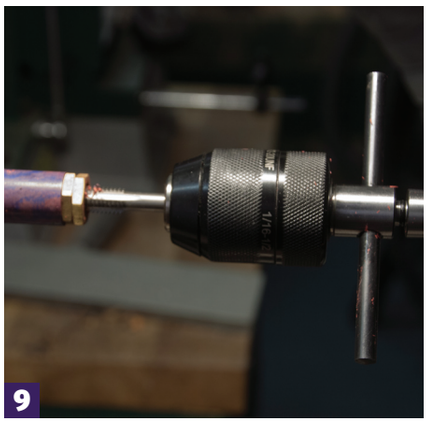
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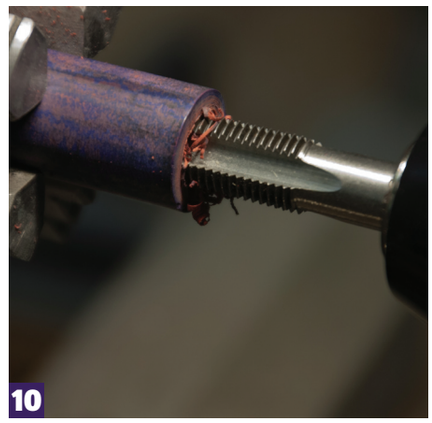
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9 Here you can see the drill chuck tapping attachment mounted in the tailstock. Once carefully aligned (my tailstock has a lot of play against the lathe bed) this makes accurate tapping an easy task. Note the brass nuts still in place for support during tapping

10 The drilling and tapping work on the barrel is now complete and we can begin on the cap. This is mounted in the chuck, drilled through for the 8mm tip thread and drilled in steps to accommodate the writing tip, then finally drilled out to size and depth for the ¼in BSP cap to body thread. This thread is cut and then the work is reversed in the chuck and tapped to 8mm for the threaded tip that holds the clip in place

11 You should now have a barrel into which the writing tip, spring and refill fit and operate properly, a cap that fits onto the barrel without binding against the writing tip and an offcut of Ebonite which will be used to make the tip of the cap. Test the fit of the components and make any minor adjustments needed to ensure everything works properly and there is a good alignment of the pattern in the Ebonite. This can be adjusted by carefully sanding the end of the cap until properly aligned

12 The remaining piece of Ebonite is now mounted in the chuck and two spigots cut to match the external diameter of your chosen clip and for the 8mm thread

13 Cut the threads right up to the face of the work by reversing the die in the holder. There is no need for a recess behind the threads as the clip will provide the necessary clearance. This part can then be parted off using a thin parting tool. Take care not to lose this small component by holding it lightly with one hand while parting off, but do keep your fingers clear of the very unforgiving engineers' jaws

14 To shape the body of the pen, it must be mounted on the lathe using some form of mandrel. I used a standard adjustable pen mandrel with the rod altered to the internal length of the barrel, supported at the open (threaded) end by a cone bushing and at the tailstock end with a small revolving centre. Use sharp tools and take light cuts to avoid the need for unnecessary and potentially damaging tailstock pressure

15 The cap may be mounted similarly or simply mounted between 60° centres and turned near to the final dimensions

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